

AKSHANT SHARMA

Engineer · Distributed Systems · AI Infrastructure

Bengaluru, India · +91 9592948889 · Akshant Sharma · [Akshant3@gmail.com](mailto:akshant3@gmail.com) · [Linkedin](#) · [Github](#) · [Profile](#) · [Blogs](#)

SUMMARY

Staff Engineer with 7.5 years building the infrastructure layer underneath — event pipelines, storage engines, and distributed systems at load most tools weren't designed for. Contributed to **Agoda's** WhiteFalcon TSDB (1.5T events/day, Rust + Kafka + Ceph), led two engineering teams at **Wayfair** shipping a real-time global pricing engine across 250k+ SKUs per supplier, built geospatial tracking at 5k events/sec at **Delivery Hero**, and architected 7M+ sensor ingestion at **Walmart Labs** on Azure IoT Hub. Currently building open-source AI inference observability tooling in Rust.

TECHNICAL SKILLS

Languages: Java (High-Concurrency Expert) · Rust · Go · Scala · Python

Distributed Systems: Kafka (internals, partition strategy, backpressure) · Ceph · Redis · ClickHouse

Storage & Data: BigQuery · PostgreSQL · MongoDB · Azure Cosmos DB · Parquet / S3

Cloud & Infra: Kubernetes · Terraform · Helm · AWS EKS · GCP · Azure IoT Hub

Observability: OpenTelemetry · Prometheus · Grafana · ELK Stack · Distributed Tracing

AI Inference & Tools: LLM inference pipelines · AI observability · Cursor · GitHub Copilot · Claude

CURRENT PROJECT

[infra-ai-streaming](#) · Open Source · *In Progress*

Rust ingestion engine → Kafka → Go consumer → ClickHouse → Grafana

- High-throughput AI inference observability pipeline for LLM metrics (latency, token cost, anomaly detection) built for the metric cardinality and event volume that standard monitoring tools break under.
- Axum HTTP server, WAL durability, per-tenant rate limiting, circuit breaker with Redis overflow buffer, Z-score anomaly detection, Helm + HPA scaling on Kafka consumer lag. Target: 1M events/min at <100ms P99.

PROFESSIONAL EXPERIENCE

Wayfair · Bengaluru Nov 2024 – Mar 2026

Sr. Software Engineer III · Led PAS & Pricing Promotion Teams

- **Led PAS and Pricing, Promotions & Discounts teams** — owning end-to-end delivery across both product areas simultaneously; architected a GCP event-driven Global Pricing & Promotion Engine reducing price propagation from hours to sub-seconds across 20k+ suppliers.
- **250k+ SKU updates per supplier in near real-time:** Engineered a high-throughput bulk processing framework (99.99% availability) with distributed rate limiting and circuit breakers, eliminating degradation and cascading failures under peak global load.

Agoda · Bangkok Apr 2024 – Sep 2024

Sr. Software Engineer · Core Infrastructure · WhiteFalcon TSDB

- **1.5 trillion events/day TSDB:** Contributed to WhiteFalcon, Agoda's in-house Rust + Scala TSDB on Kafka — higher ingestion cardinality than any off-the-shelf TSDB supports; extended query engine for cross-tier Redis/S3 queries, solving the quantile merge problem to enable correct P95/P99 across arbitrary time ranges.
- **Cardinality-safe Kubernetes indexing + 15–20% storage reduction:** Extended the RoaringBitmap inverted index to cover K8s dimensions (modifying Rust ingestion + series ID generation) to prevent index explosion; migrated cold-tier Parquet compression from Snappy to Zstd with <1% read latency impact.

Delivery Hero · BerlinJun 2022 – Jul 2023

Sr. Software Engineer · Global Logistics Platform

- **1M+ daily orders · 5k map adjustments/sec:** Scaled logistics platform to 1M+ daily orders on AWS EKS (10k+ concurrent requests, zero downtime); built end-to-end rider tracking using OSRM processing 5k+ real-time route updates/sec.
- **Cascading failure elimination:** Designed async SQS + Kinesis pipeline decoupling order processing from notifications, guaranteeing sub-second delivery status updates for millions of active users at peak.

BrowserStack · MumbaiMay 2021 – Dec 2021

Sr. Software Engineer · Device Instrumentation

- **Reverse-engineered APKs/IPAs** to hook into native iOS/Android OS APIs, enabling fully automated FaceID/TouchID testing on remote physical devices via Camera Image Injection and Biometrics instrumentation.

Walmart Labs · BengaluruAug 2018 – May 2021

Software Engineer II · WeloT SmartBuildings Platform · 3 years

- **7M+ sensors, tens of millions of telemetry points/min:** Architected WeloT SmartBuildings ingestion layer on Azure IoT Hub; engineered real-time HVAC control loops via Azure Stream Analytics automating energy optimisation across 50+ global facilities.
- **Fault-tolerant edge-to-cloud OTA firmware framework** — reliable config syncs and updates for millions of distributed devices under intermittent network conditions.

Integration Wizards · BengaluruMar 2017 – Aug 2018

IoT Lead · Industrial IoT Platform

- **Industrial IIoT platform for Dover USA:** Ingestion backbone for oil well and compressor telemetry with 100% data delivery in low-bandwidth environments; edge-compute preprocessing to filter and aggregate locally before cloud transmission.

EDUCATION

B.E. Computer Science · Chandigarh UniversityGPA 7.2 / 10

Diploma, Electrical Engineering · CCET Chandigarh

CERTIFICATIONS

Microsoft Certified: Azure Fundamentals · Employee of the Year — Integration Wizards